



Review

A systematic review on how educators teach AI in K-12 education

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ABSTRACT

Developing Artificial Intelligence (AI) education in K-12 contexts, i.e., teaching students about AI, is critical to promote students' AI literacy. However, the state-of-the-art of AI education is not clear enough. To this end, this study reviewed 45 high-quality empirical studies on K-12 AI education over the past decade from both research and instruction perspectives. Regarding the research design, this study revealed the relationship between publication year, sample size, learning stage, educational setting, research method, research focus and duration. Regarding the instruction design, this study revealed the relationship between learning stage, pedagogical strategy, learning tool, learning activity, learning content, assessment method and learning effect. Besides, this study also derived recommendations for research (i.e., time allocation, samples selection, longitudinal design, rigorous methodology and technical democracy) and instruction (i.e., group learning, authentic context, teacher involvement, triangular evidence and learning scaffolding). Overall, the main findings indicate that K-12 AI education has the potential to develop students' AI literacy, which contains AI knowledge, AI affectivity, and AI thinking. However, deficiencies in research and instructional design still remain, including short durations, small sample sizes, non-standardized research methods, lack of long-term and cross-age AI curriculum, etc. This study also discussed several critical topics for future research and instruction.

1. Introduction

Nowadays, the rapid development of Artificial Intelligence (AI) is fundamentally transforming the industries and human lifestyles. This is evidenced by the increasing use of AI to address complex problems in fields such as healthcare, engineering, economics, agriculture, shopping, transportation and psychology (Su et al., 2023a). The ubiquity of AI applications requires future generations to acquire sufficient AI literacy to most effectively use such technologies (Chiu, 2021). In this context, K-12 AI education has become increasingly essential to preparing children early for social and technological changes (Ng et al., 2023a). This section will clarify the definition of AI education, elucidate the development of K-12 AI education, and provide an overview of existing relevant reviews.

1.1. Definition of AI education

According to the diverse AI roles in education, there are multiple interpretations of AI education. Through a systematic literature review, Xu and Ouyang (2022) summarized three AI roles in education: (1) AI as a direct mediator (a platform or a tool); (2) AI as a

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supplementary assistant (data mining, learning analysis and predictive ability); and (3) AI as a new subject (an instructor or a student). Currently, AI plays the former two roles, which are dedicated to meeting demands, assessing performance or providing feedback in various educational scenarios (Marr, 2018). AIED (AI in Education) has captured the interest of educators who are devoted to enhancing the application of AI technologies in teaching, learning, evaluation, and management (e.g., Moore et al., 2023; Nguyen et al., 2023). On this basis, AI education can be broadly defined as the use of AI technology to accelerate the reform of talent training modes.

However, the possibilities for AI in education exist beyond simple online platforms, tools or applications. Especially, the emergence of ChatGPT highlights the potential for AI as a new subject to replace the original subjects (Jeon & Lee, 2023). In the future, AI may work as a tutor to transform the instructor-student relationship, a tutee to transform the student-self relationship, and a peer to transform the student-student relationship (Xu & Ouyang, 2022). In this context, a growing body of studies emphasize that future generations should acquire sufficient AI literacy to better adapt to human-machine collaboration (Chiu et al., 2021; Long & Magerko, 2020). With the rapid advances in technology, the concept of literacy has transcended its original meaning associated with books and printing (Buckingham & Burn, 2007). As an emerging technology that simulates and extends human intelligence, AI involves multiple disciplines such as neurophysiology, psychology, computer science, and mathematics, etc. The cross-disciplinary nature of AI requires both domain-specific and domain-general attributes of AI literacy (Authors, 2024a). The former enables students to master scientific, technical and ethics foundations of AI, while the latter equips students with the higher-order thinking required to address unfamiliar problems, uncertainties, or dilemmas in an AI-powered society.

Education should not only take the present situation into consideration but also be future-oriented (Huang, 2021). The escalating significance of AI literacy necessitates a shift in AI education from “AI in education” to “education about AI”. From a domain-specific perspective, this type of AI education focuses on designing learning contents within the AI discipline, such as “knowledge in AI,” “impact of AI,” and “process in AI” (Chiu, 2021). From a domain-general perspective, this type of AI education focuses on developing higher-order thinking activities (e.g., craft AI-based artifacts, design AI applications) that involve real-world problems (Ng et al., 2021). These two aspects form the meaning of “AI education” in this paper, which aims to cultivate students’ AI literacy by teaching students about AI through higher-order thinking activities in real-world contexts.

1.2. Development of K-12 AI education

Various governmental initiatives, university, school and enterprise efforts have been expended to promote K-12 AI education. For instance, UNESCO (2020) announced an education policy titled “AI in Education: Change at the Speed of Learning” to describe the current practices of AI curriculum in K-12 schools. Besides, AAAI and CSTA (2019) proposed the “five big ideas” to elucidate essential AI concepts for K-12 students in the United States in terms of perception, representation and reasoning, learning, natural interaction, and societal impact of AI. Furthermore, ISTE (2022) provided a set of guidelines for teachers who are seeking AI instructional and curriculum resources for all grades and disciplines in K-12. In alignment with the national trend, Macquarie University’s School of Education (MUSE) collaborated with IBM to propose an inclusive AI curriculum framework for middle schools (Sabuncuoglu, 2020). Besides, East China Normal University (ECNU) collaborated with SenseTime to publish the inaugural series of high school textbooks titled “Fundamentals of Artificial Intelligence” (Tang et al., 2018). Moreover, the Massachusetts Institute of Technology (MIT) focused on creating AI tools for kindergarten children - robots, and examined their uses and effects (Williams et al., 2019). Therefore, the development of AI education guidelines, curriculum, and tools has become a national strategy pushed by public and private sectors, which reflects a concerted commitment to systematically advancing educational frameworks aligned with the evolving landscape of AI technologies. Nevertheless, these curriculum and tools still require rigorous examination to better understand global guidelines and develop effective instructional design in K-12 education.

1.3. Reviews on studies of K-12 AI education

With a development of AI education, researchers have initiated empirical studies to report on AI education in K-12 schools. Several exploratory reviews have started to summarize the effective practices of AI education across different levels. For instance, Ng et al. (2023d) performed a thematic analysis of 50 studies from 2016 to 2022 to evaluate pedagogy, instructional tools, learning contents, and assessment methods of AI education in secondary school. However, their studies did not focus on the entire range of K-12 education.

Von Wangenheim et al. (2021) presented a ten-year systematic mapping of emerging visual tools from 24 artifacts to support ML teaching in K-12 settings in terms of educational characteristics, deployment, and evaluation of these tools. Additionally, Ng et al. (2023a) conducted a thematic and content analysis of 49 empirical studies from 2000 to 2020 to summarize the teaching tools and pedagogical strategies of AI education in K-12 levels. However, these reviews have not revealed what is being taught about AI and how it affects student learning in K-12 education.

Su et al. (2023a) synthesized 21 empirical studies from 1984 to 2021 to show how AI curriculum have been designed and implemented in K-12 classrooms in terms of learning goals, contents, teaching methods, assessments, and learning effects. Yue et al. (2022) analyzed 32 empirical studies from 2010 to 2022 both descriptively and thematically to provide the research status, pedagogical design, assessments and outcomes of the AI teaching programs. Sanusi et al. (2023c) reviewed 43 conference and journal papers to identify three focus areas for K-12 ML teaching: curriculum development, technology development, pedagogical development, and teacher training/professional development. These reviews summarize learning contents, tools, and pedagogy in K-12 AI education while assessing students’ learning effects, which provides valuable insights for this study. However, they involve several

limitations: (1) did not focus on studies within a refined time span; (2) did not ensure the quality of reviewed papers, as the selected papers included many conference papers; (3) did not utilize multiple search strategies to retrieve sufficient papers; (4) did not cover more AI topics such as neural networks, natural language processing, etc. (5) did not examine the research design of empirical studies; (6) did not examine the relationships between multiple coded elements. The term “AI literacy” was first introduced in 2016, which signaled the rise of AI education for non-technical students (Kandlhofer et al., 2019). Thus, the last decade is a suitable time span to explore K-12 AI education.

Considering aforementioned limitations, the purpose of this review is twofold: (1) to systematically review high-quality empirical studies from January 2014 to April 2024 on K-12 AI education from both research and instruction perspectives, and (2) to explore future directions for research and instruction in K-12 AI education based on the reviewed papers. The term “empirical study” refers to field-based research which gathers data from real-world observations or instructional experiments, such as action research, design-based research, experimental research, and case study (Flynn et al., 1990). Therefore, this review aims to examine the empirical evidence on K-12 AI education to answer three research questions.

- (1) What research has been conducted on K-12 AI education?

(2) How has K-12 AI education been designed and implemented?

(3) What recommendations for K-12 AI education are addressed in these empirical studies?

2. Method

The review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021) for conducting a systematic literature review and proceeds through four stages: data collection, data selection, data extraction, and data synthesis.

2.1. Literature search

The literature search was conducted in April 2024 using the Web of Science (WOS) and Scopus Databases. The search keywords were constructed by Boolean logic as follows: Topics = (“Artificial Intelligence” OR “Machine Learning” OR “Deep Learning” OR “Neuralnetwork*” OR “Natural Language Processing”) AND (Education OR School OR Curriculum OR Curricula OR Course OR Learn* OR Teach* OR Instruction OR Pedagogy OR Student) AND (“K12” OR “K 12” OR “high school” OR “secondary school” OR “middle school” OR “primary school” OR “elementary school” OR Child* OR Kindergarten OR “Preschool”). The initial search yielded 20738 WOS publications and 16826 Scopus publications.

2.2. Screening and data extraction

- The selection process involved two stages.

Stage 1. The first stage was a preliminary screening.

Publications retrieved from the query were considered if they were.

(1) published between January 2014 to April 2024.

(2) written in English, but not necessarily in an English-speaking country.

(3) research areas categorized as “Education Educational Research” in WOS or “Social Science” in Scopus.

(4) article or journal paper or journal article or research article (including early access).

(5) sourced from Web of Science Core Collection in WOS or Scopus CiteScore Q1 or Q2 in Education.

After the preliminary screening, 875 WOS publications and 429 Scopus publications were yielded for further selection. After removing duplicates, 1093 articles were retained.

Table 1
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Papers belonged to empirical studies.	Review, editorials, or other theoretical papers.
Papers reported studies conducted in K-12.	Papers focused on college/university education or adult education.
Papers in which the participants were students.	Papers in which the participants were teachers.
Papers involved in teaching and learning AI.	Papers that mentioned the term “AI” but were actually about applying AI in education or investing participants’ attitudes, intentions, or perception of AI education without implement details.

Stage 2. The second stage applied the following selection criteria (See Table 1) to identify which papers could be included in this review.

The titles and abstracts of 1093 papers were analyzed according to the above four inclusion criteria. As a result, 60 papers remained. Subsequently, all papers were full-text read by two back-to-back authors. 15 papers were eliminated because two studies investigated teachers' evaluations of AI learning tools developed in K-12 education (Park & Shin, 2021, 2022), four studies involved the application of AI tools in early childhood education (Dujčić Rodić et al., 2023; Kewalramani et al., 2021; Tian & Cui, 2022; Wang et al., 2021), and nine other studies focused on the development of K-12 AI learning tools (Cen et al., 2023; Naya-Varela et al., 2023), curriculum (Bellas et al., 2023) or assessment (Gresse Von Wangenheim et al., 2022); students' intentions and perceptions of learning AI (Chai et al., 2021; Sing et al., 2022); or the influencing factors (e.g., AI knowledge, AI readiness, AI engagement and AI attitude) of AI education (Huang et al., 2024; Xia et al., 2023a, 2023b), but did not provide details on the implementation of AI education. In the end, 45 papers were left.

2.3. Snowballing searching

To track papers that we might have ignored initially, we utilized the snowballing approach (Wohlin, 2014) to examine the references and citations of well-referenced literature reviews and previously selected papers based on our predetermined eligibility criteria (see Table 2).

In the first round of snowballing, three well-referenced literature reviews (Gresse von Wangenheim et al., 2021; Ng et al., 2023a; Sanusi et al., 2023c) were selected as seeds to search for papers in Web of Science. 351 references from backward snowballing and 63 citations from forward snowballing were found (414 papers in total). One new paper was selected based on the above four inclusion criteria.

In the second round of snowballing, four papers (Chiu et al., 2021; Estevez et al., 2019; Salas-Pilco, 2020; Xia et al., 2022) that were highly cited in the aforementioned search were identified as new seeds. 206 references from backward snowball and 155 citations from forward snowball were examined (361 papers in total). As a result, two new papers were singled out.

In the third round of snowballing, two papers (Chang & Tsai, 2021; Kong et al., 2023) were identified as new seeds. 157 references from backward snowball and 12 citations from forward snowball were examined (169 papers in total), but no more papers were found. This indicates that the iterative process of the snowball approach could be ended.

Therefore, after employing 3 rounds of snowballing searches, two additional relevant paper was identified. Finally, 47 papers were included for further coding. An overview of the search protocol is presented in a PRISMA diagram (see Fig. 1).

2.4. Quality appraisal

The quality of the papers was evaluated following the criteria employed by Gao et al. (2023), which were developed based on the evaluation of qualitative studies (Savin-Baden & Major, 2007) and quantitative studies (NICE, 2012). Among the included studies, there are qualitative studies ($n = 16$), quantitative studies ($n = 10$), and mixed method studies ($n = 21$). The quality of mixed-method studies was evaluated according to both qualitative and quantitative criteria. As shown in Table 3, paper quality was measured from “no mention” (NM, 0), “some mention” (SM, 1), “good mention” (GM, 2) to “extensive mention” (EM, 3). Each paper was assigned an average score based on its results in each category. The first and second authors evaluated three papers based on the criteria. All three papers scored above 2 (good mention), but for slightly different reasons. After reaching consensus on the criteria through discussion, the first author performed the quality appraisal of the remaining papers. Finally, 45 papers were included in the analysis except for two papers (Chiang et al., 2023; Zammit et al., 2022) that scored below 2.

Table 2
Summary of snowballing approach.

Procedure	Seeds	References	Selected	Citations	Selected
Round 1	Gresse von Wangenheim et al. (2021)	94	0	25	0 (+3 ^a)
	Ng et al. (2023a)	173	1 (+3 ^a)	23	0 (+2 ^a)
	Sanusi et al. (2023c)	84	0 (+1 ^a)	15	0 (+1 ^a)
Round 2	Chiu et al. (2021)	47	0	43	0 (+6 ^a)
	Estevez et al. (2019)	46	0	43	1 (+4 ^a)
	Salas-Pilco (2020)	54	0	20	0
	Xia et al. (2022)	59	0 (+1 ^a)	49	0 (+6 ^a)
	Chang & Tsai (2021)	86	0	6	0
Round 3	Kong et al., 2023	71	0 (+2 ^a)	6	0 (+2 ^a)
Total		714	1	230	1

Note.

^a The number of papers selected before, which are not added to the total.

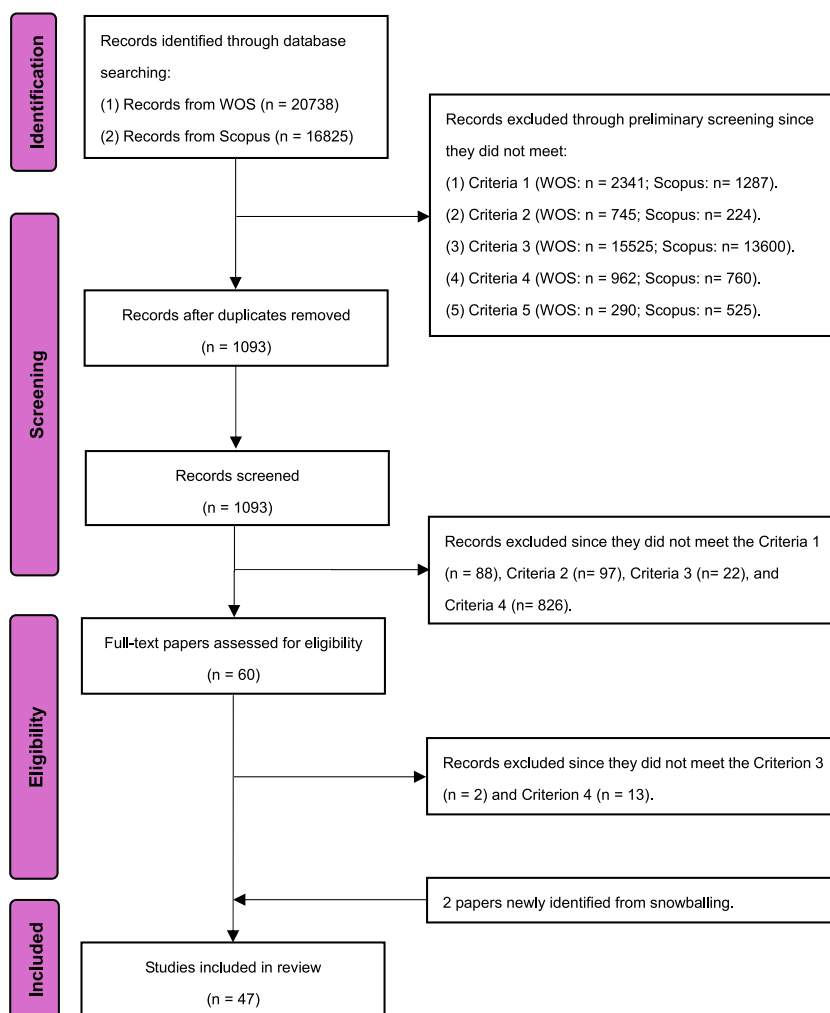


Fig. 1. PRISMA diagram.

Table 3
Criteria for the quality appraisal.

Criteria		NM (0)	SM (1)	GM (2)	EM (3)
Qualitative studies	Study is clear methodologically	0	0	16	21
	Study theoretically situated	0	5	19	13
	Ethical process transparent	19	3	9	6
	Researcher(s') relation to participants is clear	0	0	19	18
	Researcher(s') relation to the data is clear	0	1	18	18
	Researcher(s') takes a critical stance towards own research	0	0	29	8
	Congruence between methodology and methods used for data collection, analysis, and interpretation	0	3	18	16
	Participants involvement in data interpretation	0	5	15	17
	Limitations voiced	9	2	21	5
	Is the source population or source area well-described?	0	2	16	13
Quantitative studies	Were interventions and comparisons well-described and appropriate?	0	6	9	16
	Were outcome measures reliable?	0	6	15	10
	Were outcomes relevant?	1	1	12	17
	Were the analytical methods appropriate?	0	4	17	10
	Are the study results internally valid (i.e., unbiased)?	0	2	26	3
	Are the findings generalizable to the source population (i.e., externally valid)?	0	4	24	3

Note. The above criteria for the quality appraisal were derived from Gao et al. (2023).

2.5. Coding scheme

General information about 45 papers including publish time, learning stage, educational setting, and paper title were coded into Appendix A. The specific coding elements for each research question are as follows.

2.5.1. Coding elements for Q1

To answer the first question, information on sample size, duration, research method, research type, analysis methods and research focus were coded in Appendix B. By synthesizing Lv et al. (2023) and Authors (2018, 2023), sample sizes were categorized as 4–20, 21–40, 41–60, 61–80, and >80; durations were categorized as ≤1 day, ≤1 week, ≤3 weeks, ≤5 weeks, ≤9 weeks, and ≥12 weeks; and research methods were categorized as true experiment, quasi-experiment, non-experiment, case study, and design-based research. Regarding the research types, the first three categories of research methods will be denoted by specialized symbols. Analysis methods will be determined based on the specific adoption of each study. Adapted from Sanusi et al. (2023c)’s summary of research focus in K-12 ML teaching and learning, four elements were identified to determine the research focus of each study: (1) pedagogical development: strategies employed to teach AI; (2) curriculum development: content designed to introduce AI; (3) tool development: technology developed to teach AI; and (4) evaluation development: approaches employed to evaluate AI learning.

2.5.2. Coding elements for Q2

The rise of learning tools and pedagogical strategies has attributed the popularity of AI teaching (Ng et al., 2023a; Xia et al., 2022). Incorporating AI education into schools also requires a comprehensive evaluation of students’ learning effects (Kim & Kwon, 2024a). According to the definition of AI education in Section 1.1, the learning effects of AI education focus on AI literacy, which was inextricably linked to the design of learning contents and activities. On this basis, a set of instructional elements based on Scott’s (2007) model of curriculum design was identified, including learning tool, pedagogical strategy, learning activity, learning content, assessment method, and learning effect (see Appendix C). Meanwhile, learning stages were also incorporated into the framework to observe the specifics of the above elements in K-12 education. Fig. 2 summarizes the theoretical framework for coding.

● Learning supports

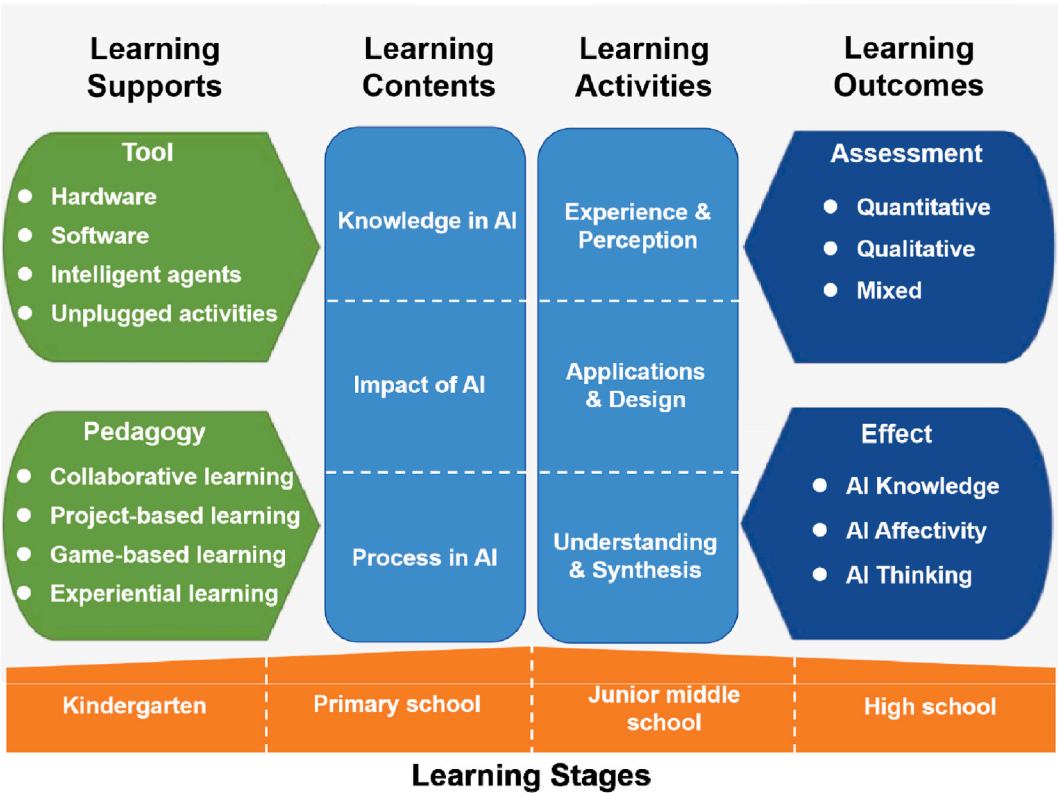


Fig. 2. The theoretical framework for designing and implementing AI education
Note. The dimensions of the inductive thematic analysis were based on Ng et al. (2023a, 2023b, 2023d), Lv et al. (2023) and Authors (2023, 2024a, 2024b).

Four categories of learning tools aligned with Ng et al. (2023a, 2023d) were identified, including hardware (e.g., robotics, sensors and Arduino devices), software (e.g., programming environments, WebAPPs), intelligent agents (e.g., machine learning trainers, expert systems, chatbots) and unplugged activities (e.g., lectures, textbooks, storytelling).

Four categories of pedagogical strategies were identified by integrating Ng et al. (2023a, 2023d), including (1) Collaborative learning: students solve problems, accomplish tasks, or create products through group intelligence (Van Leeuwen & Janssen, 2019); (2) Project-based learning: students acquire knowledge and skills by investigating authentic problems and developing real-world products (Chen & Yang, 2019); (3) Game-based learning: students enhance their understanding, interest, motivation and engagement by incorporating game elements (Sun et al., 2023b); and (4) Experiential learning: students learn by doing through hands-on learning experiences (Chiu et al., 2021).

● Learning contents

Three types of learning contents adapted from Chiu (2021) were identified, including (1) Knowledge in AI: definition, history, application and latest development of AI; (2) Impact of AI: social impact, AI ethics and human bias; and (3) Process in AI: principles of how AI use data to develop their abilities, which include modeling, statistical inference, and algorithms, etc.

● Learning activities

Three levels of AI learning activities derived from Authors (2024b) were identified, including (1) Experience & Perception: students experience simplified or pre-trained models to produce artifacts using block-based programming; (2) Applications & Design: students go through the complete model training process to produce simple AI systems using low-code programming; and (3) Synthesis & Creation: students produce multi-modal AI systems by combining different tools, algorithms and models.

● Learning outcomes

Three categories of assessment methods aligned with Ng et al. (2023d) were identified, including quantitative, qualitative and mixed methods.

Regarding the learning effects, three dimensions of AI literacy divided by Authors (2024a) were identified, including (1) AI knowledge: basic concepts and techniques behind AI; (2) AI affectivity: perception on the relationship between AI, humans and society; and (3) AI thinking: high-order thinking required to apply AI to solve unfamiliar problems, uncertainties, or dilemmas. This categorization is based on Anderson & Krathwohl (2001)'s revised Bloom's Taxonomy of Educational Objectives (BTEO), which emphasizes cognitive development by integrating knowledge and thinking along the continuum from recognition to creation. Additionally, the early BTEO included an affective domain, which encompassed interests, attitudes and values along the continuum from acceptance to characterization (Krathwohl et al., 1964). Moreover, the affective and cognitive continuums overlap significantly, which contributes to the overall development of the individual. On this basis, AI knowledge, AI affectivity, and AI thinking were identified as the three dimensions that logically constitute AI literacy.

2.5.3. Coding elements for Q3

To answer the third question, research and instructional recommendations based on Authors (2023) were generalized from findings, implications, limitations and future directions in these empirical studies (see Appendix D). This will not only help teachers implement effective K-12 AI instruction, but also encourage researchers to conduct more valuable and standardized research on K-12 AI education.

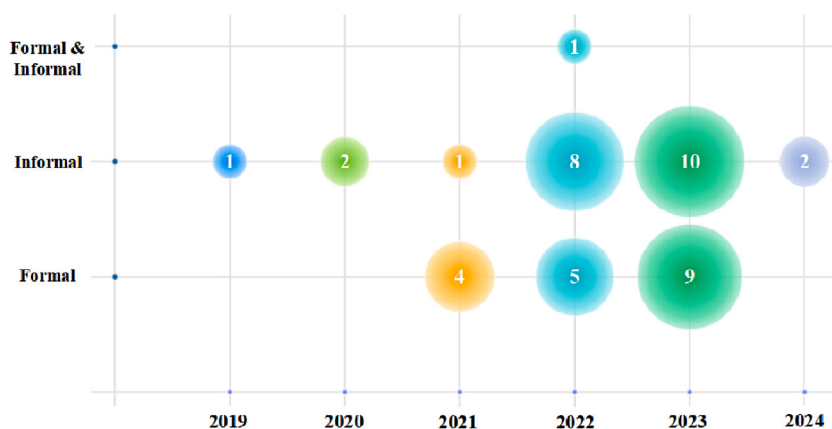


Fig. 3. Publishing year and educational setting.

Note. Since two studies (Lin et al., 2023b; Priya et al., 2022) did not mention their settings, 43 samples were included in the statistics.

For Q1 and Q2, inductive thematic analysis was employed to categorize content with similar meanings into themes pertaining to above elements. All papers were qualitatively coded by the first and second authors, which demonstrated a high inter-reliability (Cohen's Kappa = .80). Discrepancies in responses between the two researchers were subsequently compared and discussed. For Q3, the first author employed an inductive thematic analysis to code and categorize the recommendations into research or instructional themes.

3. Results

In this section, Sankey diagrams were used to visualize the main results. These diagrams represent categories with lines whose widths are proportional to flow magnitude, thereby illustrating flow distribution and the relative sizes of different categories (Vosough et al., 2019).

3.1. What research has been conducted on K-12 AI education?

This section, analyzed the 45 studies from the perspectives of publishing year, sample size, learning stage, educational setting, duration, research method and research focus, as well as the intersecting relationship between any two of them.

3.1.1. The rising status of K-12 AI education

According to Fig. 3, the publication year of K-12 AI education has been concentrated in 2019–2024 in the last decade. Among them, the number of relevant papers published in 2022 and 2023 accounted for 76.7% ($n = 33$) of the total publications.

A further mapping between publishing year and educational setting was performed. It was identified that all papers between 2019 and 2020 were implemented in informal settings (e.g., workshops, after-school programs, summer camps), which indicates that early researchers tended to explore the effects of K-12 AI education in informal settings. Research on teaching AI in school curriculum has gradually increased since 2021, indicating the rising status of K-12 AI education. This trend is likely to be more pronounced as the literature search is still in early 2024. Besides, there was also a study (Park et al., 2022) conducted separate experiments in formal and informal settings, respectively.

3.1.2. Different learning stages vary in duration and sample size

As shown in Fig. 4, junior middle schools (15 or 35.7%) were the most researched educational levels, followed by primary schools (13 or 31%) and high schools (11 or 26.2%), and finally kindergartens (3 or 7.1%). This indicates that research on K-12 AI education is primarily conducted in junior middle schools.

Besides, most high school experiments lasted less than 3 weeks (8 or 72.7%) and had fewer than 40 participants (6 or 54.5%). Most

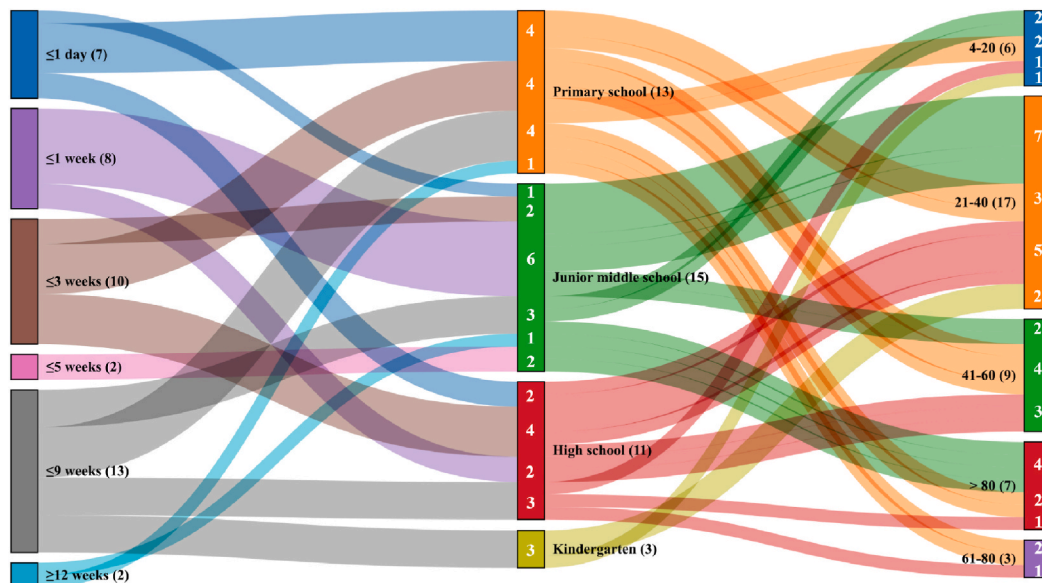


Fig. 4. Duration and sample size corresponding to each learning stage.

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023; Vartiainen et al., 2020) did not specify the learning stage, 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Since Huang & Qiao (2022) and Martins et al. (2023)'s study did not specify the sample size of the two learning stages involved, they are excluded from the statistics. Therefore, 42 samples were included in the final statistics. Besides, some studies (Dai et al., 2023b; Kong et al., 2023; Relmasira et al., 2023; Williams et al., 2023; Xia et al., 2022) conducted multi-stage experiments, and only the largest sample size participating in the experiment was included in the analysis.

junior middle school experiments lasted less than 5 weeks (11 or 73.3%) and had fewer than 60 participants (11 or 73.3%). Most primary school experiments lasted for less than 9 weeks (12 or 92.3%) and involved fewer than 60 participants (9 or 69.2%). All kindergarten experiments lasted for less than 9 weeks with less than 40 participants. Meanwhile, experiments lasting more than 12 weeks were conducted in junior middle (1 or 50%) and primary school (1 or 50%) with more than 80 participants. This indicates that higher-grade experiments are shorter than lower-grade experiments, and large-scale experiments were concentrated in primary and junior middle schools.

3.1.3. Different research methods vary in duration and sample size

As illustrated in Fig. 5, the most popular research method was quasi-experiment (19 or 42.2%), followed by case study (13 or 28.9%). Meanwhile, 14 of the 19 quasi-experimental studies were found to have no control group.

Besides, studies with sample sizes greater than 60 were predominantly quasi-experiments (6 or 54.5%), which usually lasted for more than 3 weeks (11 or 57.9%). Most case studies (11 or 84.6%) and design-based research (6 or 100%) had sample sizes of less than 60, with 14 or 73.7% out of them lasting less than 3 weeks. This indicates that the experimental research typically requires larger samples and longer duration than other types of research.

3.1.4. Different research focuses vary in learning stages and research methods

According to Fig. 6, the most popular research focus was pedagogical development (19 or 41.3%) and curriculum development (19 or 41.3%), followed by tool development (8 or 17.4%), and no studies focused on evaluation development.

Besides, most pedagogical development studies were quasi-experimental (6 or 31.6%) or case studies (6 or 31.6%) and conducted in primary schools (8 or 42.1%). Most tool development studies were case studies (3 or 37.5%) or design-based research (2 or 25%) and conducted in high schools (4 or 50%). Most curriculum development studies were quasi-experiments (12 or 63.2%) and conducted in junior middle schools (10 or 52.6%). This indicates that the lower grades focus on how to get students to effectively learn introductory AI (i.e., developing pedagogy), while the higher grades focus on getting students to learn systematic AI knowledge (i.e., developing curriculum) and experience AI practices (i.e., developing tools). Besides, pedagogical and curriculum development studies prefer quasi-experiments, while tool development studies favor case studies.

3.2. How has K-12 AI education been designed and implemented?

This section analyzed the 45 studies from the perspectives of learning stage, pedagogical strategy, learning tool, learning activity, learning content, assessment method, and learning effect, as well as the intersecting relationship between any two of them.

3.2.1. Different learning activities vary in learning stages and pedagogical strategies

As shown in Fig. 7, most learning activities focused on Experience & Perception (54 or 65.9%), followed by Application & Design (21 or 25.6%), and finally Synthesis & Creation (7 or 8.5%). This indicates that the overall difficulty of K-12 AI learning activity is still at the introductory level. Besides, the most adopted pedagogy in K-12 AI education was collaborative learning (33 or 40.2%), followed by project-based learning (28 or 34.1%), and finally experiential learning (13 or 15.9%) and game-based learning (8 or 9.6%).

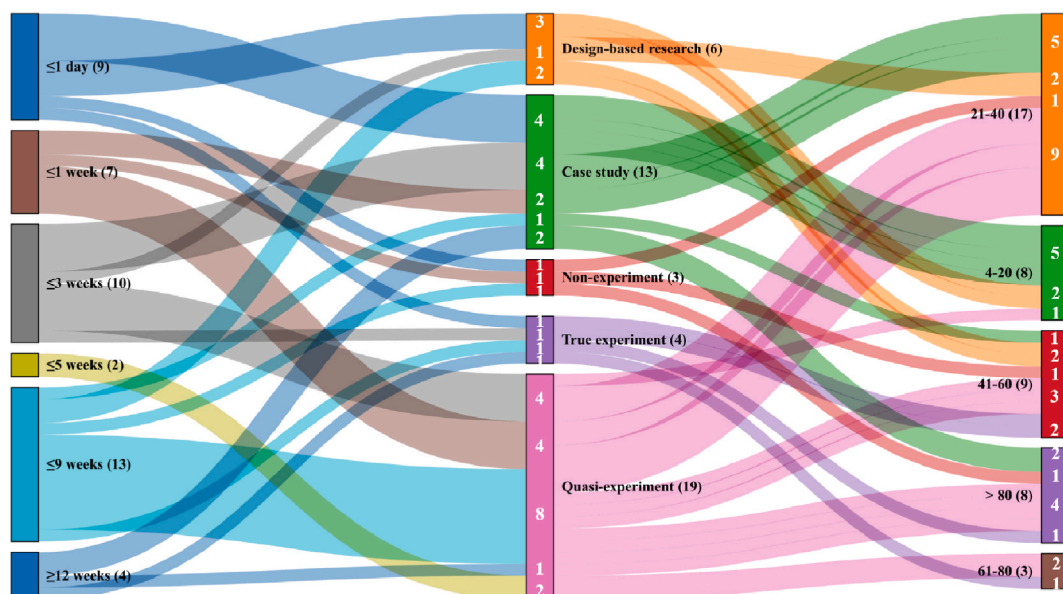


Fig. 5. Duration and sample size corresponding to each research method.

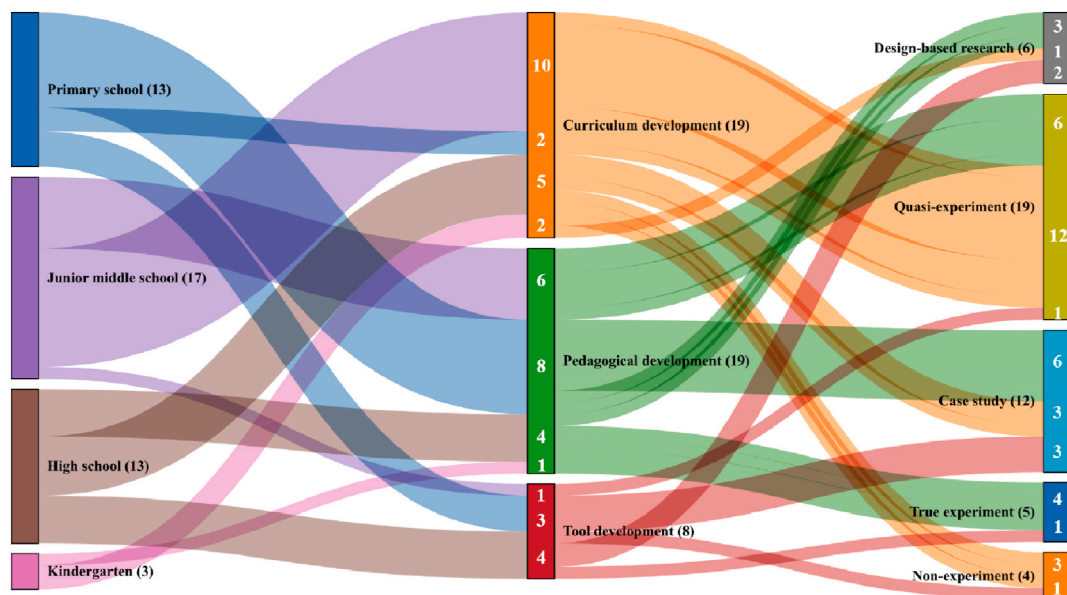


Fig. 6. Research methods and learning stages corresponding to each research focus

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023c; Vartiainen et al., 2020) did not specify the learning stage, 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Therefore, 46 samples were included in the statistics.

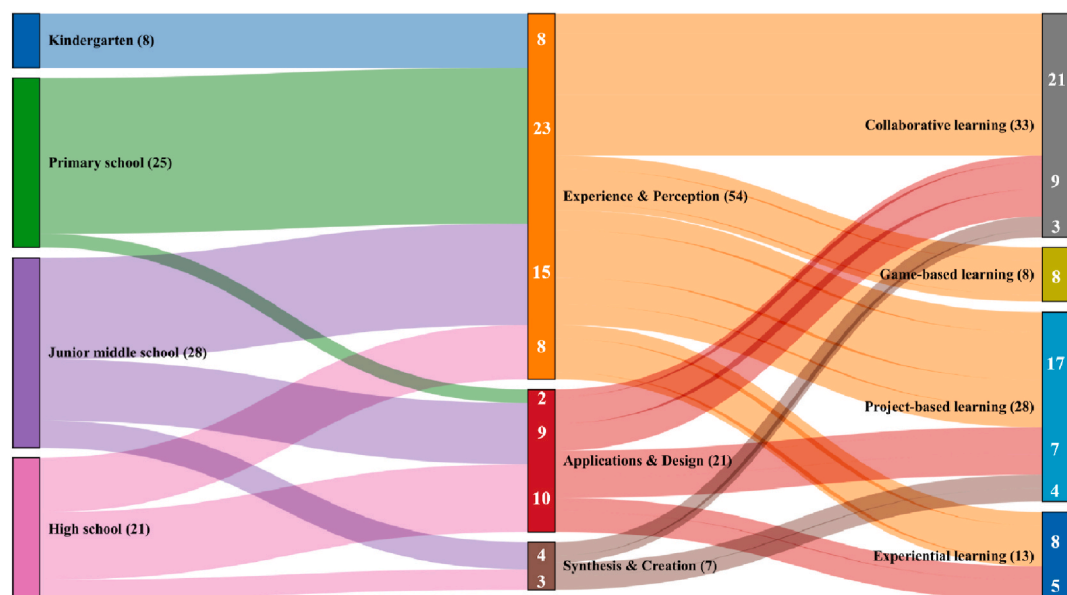


Fig. 7. Pedagogical strategies and learning stages corresponding to each learning activity.

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023c; Vartiainen et al., 2020) did not specify the learning stage, 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Besides, one study would adopt multiple pedagogical strategies. Finally, 82 samples were included in the statistics.

Meanwhile, 23 or 82.1% out of the project-based learning were employed concomitantly with collaborative learning, which illustrated the close connection between the two types of pedagogy.

Besides, most game-based learning strategies (8 or 100%) were adopted by Experience & Perception activities in kindergartens or primary schools (7 or 87.5%). Most experiential learning strategies were adopted by Experience & Perception (8 or 61.5%) activities in junior middle schools (5 or 62.5%). Applications & Design and Synthesis & Creation activities were primarily implemented through collaborative or project-based learning strategies (23 or 82.1%) in junior middle or high schools (21 or 91.3%). Consequently, game-

based and experiential learning strategies are commonly used in introductory activities for lower grades, while collaborative and project-based learning strategies are prevalent in advanced activities for higher grades.

3.2.2. Different learning contents vary in learning activities and stages

As shown in Fig. 8, most learning contents focused on process in AI (40 or 38.1%), followed by impact of AI (34 or 32.4%), and finally knowledge in AI (31 or 29.5%). This indicates that K-12 AI education prioritizes learning “how AI works” over “what AI is”.

Besides, most kindergartens (3 or 100%) and primary schools (9 or 88.9%) taught “process in AI” contents through Experience & Perception activities. Most junior middle and high schools taught “process in AI” contents through Experience & Perception (12 or 42.9%) or Application & Design (12 or 42.9%) activities. Since the “process in AI” contents were rarely taught through Synthesis & Creation activities, this indicates that K-12 AI education prioritizes experiential learning of complex contents.

3.2.3. Different learning tools vary in learning activities and stages

According to Fig. 9, the most commonly adopted learning tools were intelligent agents (26 or 34.7%), software (23 or 30.7%), hardware (20 or 26.7%), and finally unplugged tools (6 or 8%).

Besides, most intelligent agents were adopted by junior middle and high schools (15 or 53.4%) to design Applications & Design or Synthesis & Creation activities (11 or 73.3%). All unplugged tools were used for Experience & Perception activities (6 or 100%) in primary schools (4 or 66.7%). Similarly, most hardware and software were used for Experience & Perception activities (29 or 67.4%) in primary (14 or 48.3%) and junior middle schools (10 or 34.5%). This indicates that hardware, software and unplugged tools are typically used for introductory learning activities in the lower grades. In contrast, intelligent agents are typically used for advanced learning activities in the higher grades.

3.2.4. Different learning effects vary in assessment methods and learning stages

As illustrated in Fig. 10, the most popular assessments were qualitative methods (38 or 40%), followed by quantitative methods (33 or 34.7%) and mixed methods (24 or 25.3%). Besides, most studies reported learning effects of curriculum, tools, or pedagogical interventions on students’ AI knowledge (31 or 32.6%) (e.g., AI concepts, ML concepts), AI affectivity (37 or 38.9%) (e.g., AI ethics, AI readiness, AI confidence, AI attitude), and AI thinking (27 or 28.4%) (e.g., creativity, abstract thinking, system thinking, computational thinking). Since most studies proved the positive effects of AI instruction, p-values and effect sizes were tallied in Appendix C, except for those studies that did not involve or provide relevant statistical values. Most p-values (28 or 46.7%) showed extremely significant differences (i.e., $p < 0.001$), followed (23 or 38.3%) by significant differences (i.e., $p < 0.05$). Besides, most calculations resulted in large effect sizes (18 or 37.5%), followed by small (16 or 33.3%) or medium (14 or 29.2%) effect sizes. This suggests that most AI instruction has yielded the desired results.

Besides, quantitative methods were primarily used to assess students’ AI knowledge (14 or 42.4%) in high schools (6 or 42.9%). Most qualitative methods were used to assess students’ AI thinking (16 or 42.1%) in primary (6 or 37.5%) and junior middle schools (6 or 37.5%). Most mixed methods were used to assess students’ AI affectivity (15 or 62.5%) in junior middle schools (6 or 40%).

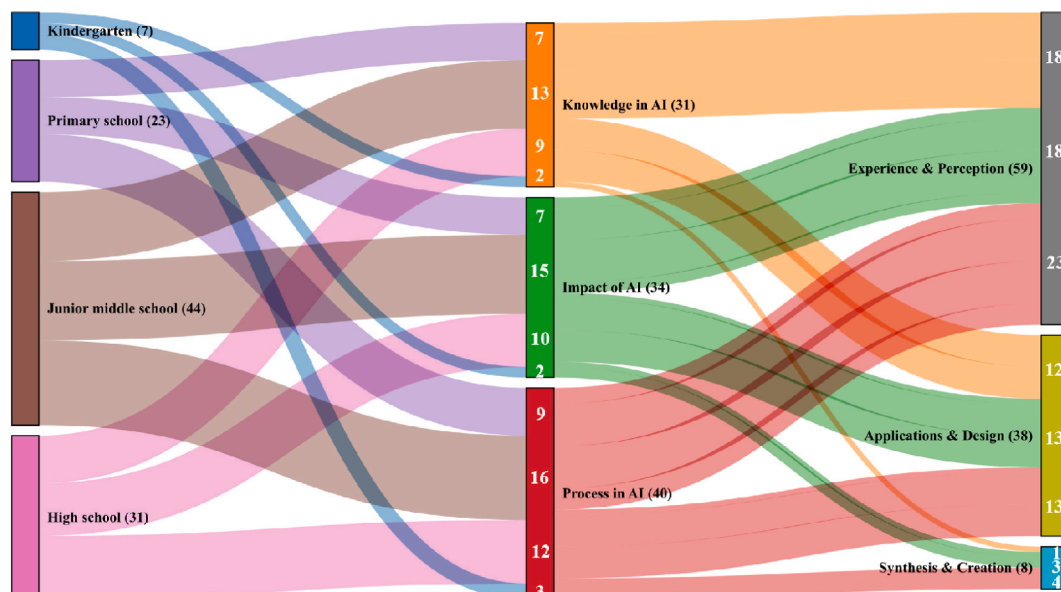


Fig. 8. Learning activities and stages corresponding to each learning content.

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023c; Vartiainen et al., 2020) did not specify the learning stage, 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Besides, one study would teach multiple types of contents. Finally, 105 samples were included in the statistics.

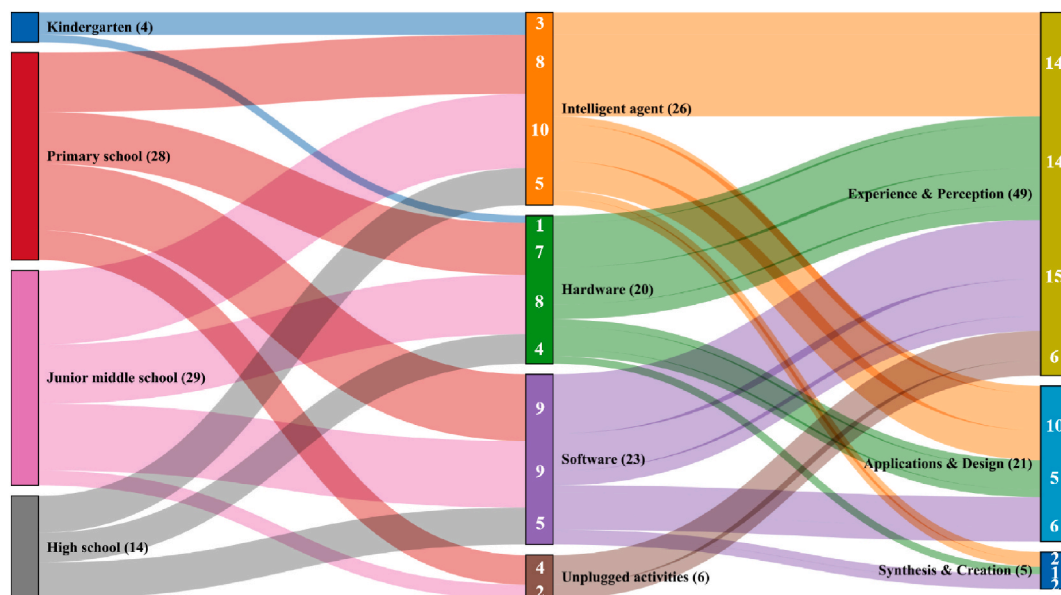


Fig. 9. Learning activities and stages corresponding to each learning tool.

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023c; Vartiainen et al., 2020) did not specify the learning stage, 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Besides, 2 studies (Huang & Qiao, 2022; Kong et al., 2023) did not mention the tools used and one study would use multiple tools. Finally, 75 samples were included in the statistics.

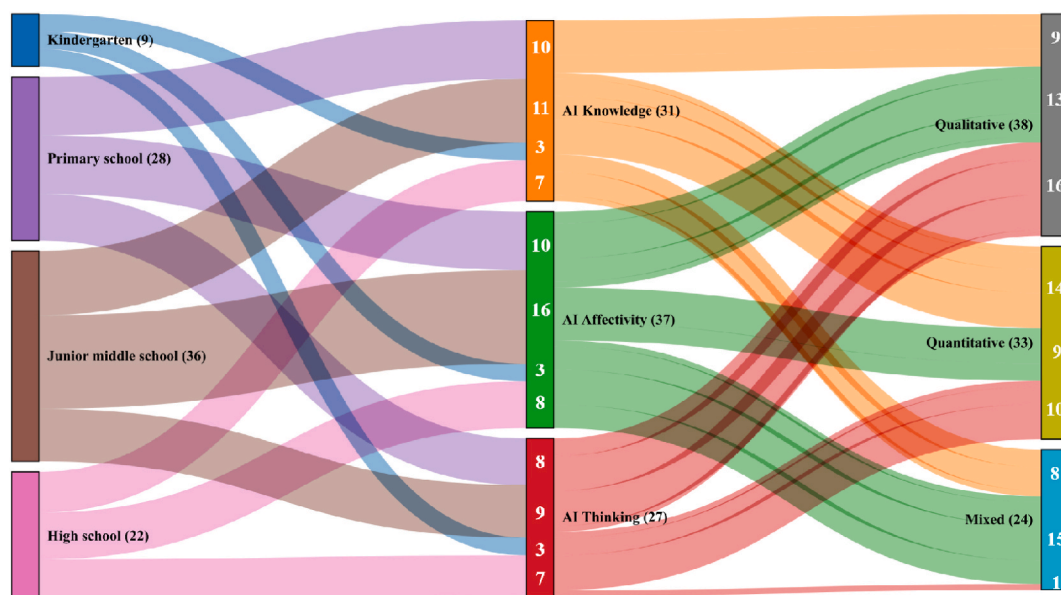


Fig. 10. Assessment methods and learning stages corresponding to each learning effect.

Note. Among the 45 studies, 2 studies (Sanusi et al., 2023c; Vartiainen et al., 2020) did not specify the learning stage, 5 studies aimed to describe students' process of exploring, developing, and testing of models (Jiang et al., 2022, 2023a, 2023b), the roles of children and researchers as meta-design partners in ML education (Irgens et al., 2022a), as well as students' naive conceptions about AI concepts (Kim et al., 2023), so they are excluded from the statistics. Among the remaining 38 studies, each study would measure multiple aspects of learning effects, and 3 studies (Huang & Qiao, 2022; Lin et al., 2023b; Martins et al., 2023) involved participants in junior middle and high schools. Finally, 95 samples were included in the statistics.

Meanwhile, most kindergarten studies measure students' learning effects through qualitative methods (7 or 77.7%). Thus, quantitative, mixed and qualitative methods are suitable for assessing knowledge, affectivity and thinking respectively. Besides, the higher grades preferred quantitative methods, whereas the lower grades preferred qualitative methods.

3.3. What implications for K-12 AI education are addressed in these empirical studies?

In order to provide researchers and teachers with research and instruction references, some useful recommendations have been extracted from 45 papers.

3.3.1. Research design

● Time allocation

Seven papers concluded that setting a longer duration can improve the reliability of research results. For example, [Kong et al. \(2023\)](#) suggested that sufficient learning time are required to strengthen high school students' comprehension of AI ethics. Similarly, [Salas-Pilco \(2020\)](#) argued that providing students with adequate space and time to develop robotic models could encourage their creativity. Moreover, [Zhang et al. \(2023\)](#) emphasized that complex AI processes such as neural networks may require longer exposures and more time for students to internalize.

● Samples selection

Twenty-three papers supported that multi-source samples are key to improving the generalizability of findings. For example, [Dai et al. \(2023a\)](#) and [Relmasira et al. \(2023\)](#) suggested that further research should explore the generalizability and transferability of the pedagogy to other student populations. Likewise, [Ng et al. \(2022\)](#) and [Kim & Kwon \(2024b\)](#) planned to broaden the sample to provide generalizable results for a wider population. Moreover, [Choi et al. \(2024\)](#) and [Vartiainen et al. \(2021\)](#) argued that future research will incorporate other age groups to provide a more comprehensive understanding of AI education.

● Longitudinal design

Eight papers suggested that longitudinal studies could capture the dynamics and complexity of student learning. For example, [Chiu et al. \(2021\)](#), [Park and Kwon \(2023\)](#) deemed that tracking teaching and learning activities will capture valuable data to support learning analytics. Additionally, [Lin et al. \(2023a\)](#) argued that a longitudinal design was necessary to examine the impact of the proposed approach on higher-order thinking, i.e. creativity and metacognitive awareness in solving complex problems. Moreover, [Sanusi et al. \(2023b\)](#) and [Xia et al. \(2022\)](#) suggested that future studies could adopt a longitudinal research design and study programs offered as part of regular schooling.

● Rigorous methodology

Nine papers highlighted that rigorous experimental design is essential for promoting research and practices in early AI education. For example, [Sanusi et al. \(2023d\)](#) argued that the lack of more structured tasks or pre-post testing makes it difficult to conclude that the ML concept was learned in a robust way. Besides, [Lu & Fan \(2023\)](#) and [Zhang et al. \(2023\)](#) addressed that due to the lack of comparison groups in teaching experiments, caution must be exercised when drawing inferences from their results. Moreover, [Kim & Kwon \(2024b\)](#) and [Williams et al. \(2023\)](#) planned a randomized controlled trials with a control group to provide stronger evidence for causation.

● Technical democracy

Four papers indicated that educational research is crucial for breaking the harmful tradition of technology development and consumption without a critical lens. For example, [Irgens et al. \(2022a\)](#) argued that the awareness of technical democracy is critical to enabling youth early to think critically about social, ethical, and political issues. Besides, [Ali et al. \(2021\)](#) deemed that future studies could focus on awareness about how evolving technology influences the spread of information. Moreover, [Jiang et al. \(2023a, 2023b\)](#) concurred that future studies should stimulate productive reasoning about how we can shape the way technology functions in society.

3.3.2. Instructional design

● Group learning

Six studies regarded group learning as an effective approach to enhance students' hands-on experience in AI education. For instance, [Ng et al. \(2023c\)](#) showed that by working together in a structured and collaborative environment, students can learn from each other and develop their AI literacy, socialization, motivation, and confidence. Besides, [Jiang et al. \(2023a\)](#) deemed that students' naive concepts will gradually be transformed into scientific concepts through explaining why they employ these AI concepts, and modifying them in response to feedback. Moreover, [Hsu et al. \(2021\)](#) and [Zhan et al. \(2022\)](#) suggested that more evidence of the correlation between the learning effects and collaborative behavioral patterns should be explored.

● Authentic context

Five studies agreed that integrating authentic elements into learning experiences can be an effective strategy for enhancing students' AI literacy. For example, [Irgens et al. \(2022b\)](#) indicated that ML education cannot be separated from the social, historical, and political contexts in which people consume and produce technology. Additionally, [Jiang et al. \(2022, 2023a\)](#) claimed that modelling unstructured data from real-world datasets can help students reason about computer decision making and understand social issues about AI technologies. Moreover, [Lin et al. \(2023b\)](#) and [Ng et al. \(2023c\)](#) argued that engaging in authentic contexts with learning activities can encourage students to create AI artifacts and recognize AI ethics.

● *Teacher involvement*

Seven papers emphasize the role of teacher involvement in AI education. For example, [Irgens et al. \(2022a\)](#) considered that children should be offered autonomy in developmentally appropriate ways with adult support. Besides, [Jiang et al. \(2022\)](#) showed that teacher involvement was crucial for helping younger children reach agreement, understand unfamiliar concepts and steps when performing AI practices. Moreover, [Jiang et al. \(2023a\)](#) argued that teachers can position themselves as co-inquirers in exploring the applications of AI technology.

● *Triangular evidence*

Twelve papers highlighted triangular evidence in data collection. For example, several studies concluded that future research could gather more valuable insights for interpreting quantitative findings by including teacher interviews, as teachers are the most knowledgeable about their students and classroom dynamics ([Hu et al., 2023](#); [Huang & Qiao, 2022](#); [Sanusi et al., 2023b](#); [Su & Yang, 2023b](#)). Besides, [Kim et al. \(2023\)](#) and [Sanusi et al. \(2023d\)](#) indicated that future research could employ a variety of data collection methods, not only student videos and artifacts, but also quantitative data such as surveys or tests to identify more generalized findings.

● *Learning scaffolding*

Nine papers identified the role of scaffolding in supporting students' learning. For example, [Lu & Fan \(2023\)](#) and [Relmasira et al. \(2023\)](#) suggested that teachers can provide students with self-directed learning materials in advance and appropriate support at the right moments. Besides, [Jiang et al. \(2023b\)](#) and [Kim et al. \(2023\)](#) advocates for teachers to develop instructional and technological strategies to help students overcome misconceptions or naive conceptions. Moreover, [Schaper et al. \(2022\)](#) developed a framework for applying computational empowerment in formal educational contexts to scaffold teenagers' learning about emerging technologies.

4. Discussion

In this section, we further summarize the main ideas of 45 papers based on Section 3, and discuss the future research prospects and practice directions for K-12 AI education.

4.1. Research status of K-12 AI education

4.1.1. Publication status

Empirical research on formal K-12 AI education has increased rapidly since 2021. In the past, students were required to use syntax programming to make neural networks, intelligent agents, and classification models, which was challenging for novices. The dearth of age-appropriate tools and comprehensive curriculum is thus one of the reasons why K-12 AI education was not popular before 2021 ([Sakulkueakulsuk et al., 2018](#)). Nowadays, students can utilize block-based programming with more developmentally appropriate learning tools to develop their AI-driven artifacts ([Long & Magerko, 2020](#)). Besides, a number of tool-supported curriculum has been developed for the primary and even pre-school levels. As demonstrated by the AI4KG curriculum, which exposes preschoolers to basic AI concepts via three AI learning tools (i.e., AI for Oceans, Teachable Machine, and Quick, Draw!) ([Su & Yang, 2023a; 2023b](#)).

However, few K-12 schools consider AI courses as independent courses. Most AI courses are combined with STEAM, robotics, or maker courses, with programming as the main content and supplemented by popular science lectures and technology exhibitions. For instance, [Estevez et al. \(2019\)](#) provided students with an academic use-create-modify scaffolding to perform K-means and artificial neural network algorithms for Scratch coding. Additionally, [Hu et al. \(2023\)](#) explored how students interact with their peers in AI teaching contents of "The Use of Micro: bit Programming Tools" and "The Definition and Application of Sensors". Similarly, [Salas-Pilco \(2020\)](#) explored how learning designs for constructing robotic models can influence students' physical, social-emotional, and intellectual learning outcomes. Moreover, [Ng et al. \(2023c\)](#) integrated AI into maker education to enhance students' motivation, career interest, confidence, collaboration, and AI literacy by making AI-driven recycling bins. Therefore, AI education in K-12 schools as a whole is still in its infancy.

4.1.2. Learning stage

K-12 AI education research is largely focused on junior middle schools, which is in contrast to previous research that advocated high schools. For instance, [Kong et al. \(2023\)](#) argued that senior secondary students are particularly suitable targets for AI education because one's political and social views start shaping at this stage. Likewise, [Estevez et al. \(2019\)](#) argued that high school is a great time to expose students to AI curriculum, as students are mature enough to handle complex issues. Additionally, [Lin et al. \(2023b\)](#) suggested

that high school students are better suited for the AI learning context because they may be interested in continuing their future professional development. Furthermore, Jiang et al. (2022) aimed to plant intellectual seeds of AI at the high school level so that they can be nurtured and grown as they enter higher education. As the concept of "intellectual seeds" suggests here, planting the seeds of AI literacy at an early age will not only help them better adapt to the AI society by acquiring knowledge and skills to address complex problems, but also inspire them to further pursue AI in their studies and careers by stimulating their interest in AI. Thus, student characteristics and developmental readiness determine high school as the optimal stage for AI education.

However, high school studies have a shorter duration and a smaller sample size than other learning stages, possibly due to the greater pressure of taking entrance examinations, as inferred by Lv et al. (2023) and Authors (2023). Large-scale experiments can enhance the generality of conclusions (Estevez et al., 2019), while longitudinal studies can help examine long-term effects over time (Chiu et al., 2021). Thus, future research on AI education in high schools should include large-scale and longitudinal studies.

4.1.3. Research method

K-12 AI education research favors quasi-experiments and case studies. However, the prevalence of quasi-experimental designs without a control group may raise concerns regarding the robustness and generalizability of the findings (Bithell, 1994). Besides, dependence on case studies may pose replication difficulties and limited generalizability due to its heavy reliance on retrospective data (Denzin & Lincoln, 2011). Although six studies (Estevez et al., 2019; Hsu et al., 2021; Irgens et al., 2022b; Relmasira et al., 2023; Salas-Pilco, 2020; Sanusi et al., 2023a) purported to utilize design-based research (DBR) as the methodology, their implementation or presentation was not standardized.

Actually, DBR does not focus only on developing an intervention (e.g. software tools, appropriate scaffolding), but also intends to generate, modify or enhance a theory through a series of experiments (Design-Based Research Collective, 2003). According to Gravemeijer and Cobb (2006), the DBR methodology consists of the following phases: (1) Preparation: defining the initial conditions of the study; (2) Implementation: intervening using defined design principles and refining the intervention through iterations; and (3) Closure: conducting a retrospective analysis. A complete DBR not only develops new theories or develops existing ones, but also establishes design principles for solving specific problems. By integrating design, research, and practice tightly, DBR bridges the gap between theory and practice encountered in traditional research methodologies in educational settings (Anderson & Shattuck, 2012).

However, with the exception of Salas-Pilco (2020) and Relmasira et al. (2023) who examined how AI impact students' learning outcomes through three or four iterations of learning design, the remaining four studies focused only on the effects of pedagogical, tool or curriculum interventions without theoretical optimization or iterative processes. The reason may be attributed to the researchers' incomplete presentation of the DBR process or an ambiguous understanding of DBR. For instance, Estevez et al. (2019) explained that their paper reported the first iteration of a long-term effort, whose results will serve as the basis for the next experiments. Likewise, Irgens et al. (2022b) claimed that their study was the third iteration of an ongoing participatory design study. Moreover, Sanusi et al. (2023a) described DBR as "design experiment", which aims to develop students' conceptual and procedural understanding of ML through the design process. Notably, half of the purported design-based studies lasted less than a day, which was not conducive to thorough implementation of the DBR methodology. In the future, K-12 AI education is expected to implement and present authentic DBR with extended duration.

4.1.4. Research focus

Lower-grade studies emphasize pedagogy development through quasi-experiments, whereas higher-grade studies emphasize curriculum or tool development through quasi-experiments and case studies. For instance, Lin et al. (2023a) compared the effects of contextualized dilemma discussion approach based on Augmented Reality (AR) and mobile learning for developing primary students' AI ethics. Besides, Ng et al. (2023c) evaluated the effects of an AI learning course on the topic of "making an AI recycling bin (p. 9724)" in enhancing secondary students' motivation, collaboration, and AI literacy. Moreover, Jiang et al. (2022) conducted a descriptive analysis of high school youth engaging in text mining practices in StoryQ, a web-based text mining and narrative modeling platform.

As AI education proliferates into K-12 settings, it is essential to develop a broader range of free curriculum and tools tailored to different grade levels and AI concepts. Particularly for the process of tool development, while case studies can provide more detailed insights into students' perceptions of their usage and effects, comparative experiments using two or more tools are also encouraged to expose which tools work, and which do not, as well as their effects in introducing AI concepts (Sanusi et al., 2023d). Meanwhile, the theoretical underpinnings of pedagogical approaches should be further detailed so that educators can adapt and apply them in various contexts. More importantly, more research is needed on how to evaluate K-12 AI education, as comprehensive assessments can provide valuable insight into learning effects (Ng et al., 2023a).

4.2. Status of K-12 AI teaching and learning

4.2.1. Learning activities

Lower grades favor Experience & Perception activities (e.g., Dai et al., 2023a; Hu et al., 2023; Kim & Kwon, 2024b; Yang et al., 2024). For instance, Dai et al. (2023a) organized an eight-week workshop in Grade 6 that introduced AI concepts using unplugged activities in the initial half and emphasized block-based programming in Scratch in the latter half. Besides, Yang et al. (2024) conducted a two-month program for kindergartens to understand how AI can help humans solve real-world problems. Moreover, Kim & Kwon (2024b) designed a two-week program to help primary students create AI-based artifacts using tangible computing tools.

Meanwhile, higher grades favor Application & Design activities (e.g., Aung et al., 2022; Jiang et al., 2022; Martins et al., 2023; Williams et al., 2023). For instance, Aung et al. (2022) conducted a four-week workshop in Grade 7–8 to take students through basic

ML steps including data collection, training, and testing using a web-based learning platform. Besides, [Martins et al. \(2023\)](#) guided middle and high school students to develop an image classification model following the basic steps of a human-centric ML process, including data preparation, model training, performance evaluation, and prediction. Moreover, [Williams et al. \(2023\)](#) taught middle school students about creative ML techniques and how to train their own supervised ML models.

Moreover, only four studies ([Lu & Fan, 2023](#); [Oskotsky et al., 2022](#); [Sanusi et al., 2023a](#); [Schaper et al., 2022](#)) focused on Synthesis & Creation activities. For instance, [Oskotsky et al. \(2022\)](#) held a three-week program at high schools that guides students in applying AI to characterize, classify, or predict COVID-19 using different types of biomedical data (gene expression, proteomics, imaging, and clinical). Besides, [Lu & Fan \(2023\)](#) instructed high school students to build a weather prediction system that combined weather data with ML techniques. Moreover, [Sanusi et al. \(2023a\)](#) guided middle school students to develop prototypes of ML applications containing multimodal data, such as images, sound, or poses.

Considering that AI curriculum was the guaranteed way to implement AI education in schools, future educators should consider cognitive development at all stages to design longer-term and cross-age AI curriculum. For instance, AI education in primary schools should emphasize Experience & Perception activities, so that students can know and understand the basics of AI. Subsequently, AI education in junior middle schools should emphasize “Application & Design” activities, so that students can apply AI concepts and applications to different scenarios. Finally, AI education in high schools should emphasize “Synthesis & Creation” activities, so that students can synthesize AI concepts and applications to creatively solve complex problems. Thus, the three levels of activities identify the goals for students’ AI literacy at each stage, which aligns with [Ng et al. \(2023b, 2023c\)](#)’s framework for scaffolding students’ knowledge and thinking from lower to high cognitive levels. On this basis, [Authors \(2024a\)](#) analyzed and compared various components of AIL, while [Ng et al. \(2023b, 2023c\)](#) distinguished different levels of AIL. The combination of the two will facilitate adaptive evaluation of AI literacy across K-12 stages.

4.2.2. Learning contents

K-12 AI education prioritizes experiential learning of complex principles. For instance, [Sanusi et al. \(2023b\)](#) introduced an unplugged paper-based activity to junior middle school students on how computers reinforce the concept of convolutional neural networks (CNNs). Besides, [Priya et al. \(2022\)](#) designed a 3D video game that presented three ML concepts (i.e., supervised learning, gradient descent and K-nearest neighbor classification) to high school students. Moreover, [Kong et al. \(2023\)](#) explained the concepts of supervised and unsupervised learning algorithms to high school students through practical examples.

Prior studies showed that junior middle students should focus on preliminary AI principles such as machine learning, natural language processing, and Turing tests ([Chiu et al., 2021](#)), while high school students should learn more advanced AI principles such as computational game theory, agent-based modeling, probabilistic reasoning and graph theories ([Estevez et al., 2019](#); [Reyes et al., 2020](#)). Design learning for authentic problems is critical to the successful teaching of these emerging principles, which aims to translate abstract ideas into concrete ones through developing digital and physical artifacts ([Chiu, 2021](#)). Once grounded in authentic design learning, students will develop a better historical and contextual understanding of principles. Therefore, educators should prioritize engaging students in physical manipulation of tangible artifacts and sensory-motor activities to facilitate their comprehension of complex AI principles.

4.2.3. Pedagogical strategies

Lower grades favor game-based and experiential learning in introductory learning activities, while higher grades favor collaborative and project-based learning in advanced learning activities. For instance, [Lin et al. \(2023a\)](#) employed game-based learning in an AR-based game with dilemma discussions to develop primary students’ AI ethics. Besides, [Aung et al. \(2022\)](#) used experiential learning to allow middle school students to build a simple AI application and train their own ML models. Moreover, [Oskotsky et al. \(2022\)](#) adopted collaborative project-based learning to enable high school students to develop AI models capable of characterizing, classifying, or predicting COVID-19 cases.

Prior research has shown that playful and experiential learning are considered as low-barrier approaches for introducing AI concepts and maintaining motivation among novices with limited computer science backgrounds ([Long & Magerko, 2020](#); [Ng et al., 2021](#)). In contrast, collaborative project-based learning involves open-ended goals, which enables students to form teams and design projects based on their interests and experiences ([Ng et al., 2023a](#); [Yue et al., 2022](#)). For instance, [Ng et al. \(2023a\)](#) found that incorporating game elements into AI curricula aids beginners in developing AI concepts and sustaining their learning motivation, while [Reyes et al. \(2020\)](#) demonstrated that experiential activities offer students hands-on experience to visualize advanced concepts and construct ML models. Besides, [Deng et al. \(2021\)](#) found that collaborative learning facilitates students’ communication with peers to acquire AI knowledge and manipulate smart devices, while [Wan et al. \(2020\)](#) found that project-based learning motivates students to tackle authentic, complex problems through real-world applications. Therefore, educators can further combine these pedagogical strategies based on different contents to create an engaging learning environment that prepares students to succeed in future AI-powered world.

4.2.4. Learning tools

The reviewed studies have shown that unplugged activities can stimulate learners’ enthusiasm and motivation, as well as facilitating their conceptual understanding of AI (e.g., [Dai et al., 2023b](#); [Kim et al., 2023](#); [Park et al., 2022](#)). Hardware such as robotics kits could engage students to use physical and digital artifacts to make AI-driven applications (e.g., [Chiu et al., 2021](#); [Salas-Pilco, 2020](#); [Zhan et al., 2022](#)). Meanwhile, most studies employed visual programming languages (e.g., Scratch, SNAP! Blockly, and App Inventor) to help novices prevent errors, support recognition over recall, and provide a limited set of domain-specific instructions to reduce

cognitive load (e.g., Estevez et al., 2019; Irgens et al., 2022a; Park & Kwon, 2023). Furthermore, intelligent agents (e.g., Google CoLab, Machine Learning for Kids and Teachable Machine) were favored by many studies to develop students' technical skills, critical thinking skills, and collaboration (e.g., Jiang et al., 2023b; Oskotsky et al., 2022; Sanusi et al., 2023a).

In addition to general AI tools that are well-designed for novices to experience AI, some teaching units incorporate AI tools developed by researchers, typified by AIThaiGen, a web-based learning platform to introduce AI concepts (Aung et al., 2022) and StoryQ, a web-based text mining and narrative modelling platform (Jiang et al., 2022, 2023a, 2023b). Prior studies have demonstrated that high-tech and programming activities offer more opportunities for students to acquire high-order thinking skills than low-tech and unplugged activities (Li et al., 2022). Therefore, more low-floor and high-ceiling tools are required so that novices can easily get started and potentially engage in increasingly complex activities.

4.2.5. Assessment methods

Generally, quantitative data was collected by surveys/questionnaires, tests/examinations, or system logs (e.g., Aung et al., 2022; Chiu 2021; Huang & Qiao, 2022), while qualitative data was collected by capturing students' interviews, video/audio materials, observations, learning artifacts, drawings, written reflections, field notes, activity reports, or photographs (e.g., Hu et al., 2023; Irgens et al., 2022a). However, there are also special cases, such as quantitative or mixed analysis for data collected in a qualitative manner. For instance, Dai et al. (2023a) scored students' understanding levels based on coded interview results. Then, nonparametric statistics—Wilcoxon signed-rank test and the Wilcoxon rank-sum test—were adopted to compare the within-group and between-group differences. Additionally, Aung et al. (2022) conducted thematic analysis on students' learning artifacts, while Chang & Tsai (2021) conducted an independent sample *t*-test after assigning scores to students' learning artifacts. This highlights the flexibility between data collection and assessment methods in K-12 AI education.

Meanwhile, prior studies have suggested that quantitative methods may not fully capture the deep understanding of AI concepts, while qualitative methods may not be feasible for large-scale assessments (Denzin & Lincoln, 2011; Perrotta & Selwyn, 2020). Although many studies adopted mixed methods to gather detailed information about students' learning experiences and outcomes, only ten studies (Ali et al., 2021; Choi et al., 2024; Dai et al., 2023b; Kim & Kwon, 2024b; Lu & Fan, 2023; Ng et al., 2023c; Su & Yang, 2023a, 2023b; Williams et al., 2023; Zhang et al., 2023) triangulated the results of qualitative analyses with quantitative findings. For instance, Ng et al. (2023c) conducted artifact-based interviews to triangulate quantitative results from questionnaires and knowledge tests. Besides, Kim & Kwon (2024b) utilized a concurrent triangulation design to collect different but complementary data on the same topic to provide a solid explanation. While the learning effects of several studies have not been statistically proven, mixed methods helps to explain the results. For instance, Su & Yang (2023a) conducted parent interviews to verify improvement in children's knowledge, skills, and attitudes after the AI4KG program, although the Wilcoxon signed-rank test did not show any significant differences due to the small sample size. Besides, Lu & Fan (2023) collected students' opinions to analyze the reasons why paired *t*-test showed no significant change in their attitudes towards AI learning after participating in the teaching experiment.

Moreover, the priority of mixed methods to evaluate students' affective-related learning outcomes was observed. Obviously, evaluation of AI education is an extremely complex and abstract process, which is not only limited to the individual's external behaviors, but also contains the individual's internal psychological and cognitive processes. Therefore, relying entirely on traditional means such as surveys and tests is far from enough to accurately assess students' learning effects in AI education. In the future, educators should employ intelligent methods to evaluate and triangulate students' learning performance to improve instructional implementation.

4.2.6. Learning effects

The reviewed studies generally reported positive instances of implementing interventions on students' AI knowledge, affectivity, and thinking. In terms of AI knowledge, several studies showed a significant improvement (Cohen's $d > .80$) on students' understandings of AI concepts (e.g., Aung et al., 2022; Chang & Tsai, 2021; Dai et al., 2023a), but few studies explicitly identified which AI concepts (e.g., machine learning, computer vision, and image recognition, etc.) were enhanced. In terms of AI affectivity, an increasing body of research highlighted improvements in students' attitudes (Cohen's $d = 2.25$) (Choi et al., 2024), ethical awareness (Cohen's $d = .91$) (Dai et al., 2023a), motivation (Cohen's $f = .50$) (Huang & Qiao, 2022) and collaboration (Cohen's $d = .81$) (Ng et al., 2023). In terms of AI thinking, the reviewed studies reported significant effects of implementing interventions on students' creativity (Cohen's $d = 2.95$) (Chang & Tsai, 2021), critical thinking (Cohen's $d = .62$) (Dai et al., 2023a) and computational thinking (Cohen's $d = .94$) (Lu & Fan, 2023).

Prior studies have emphasized that successful AI education should not superficially implement learning activities such as simply demanding that students understand concepts, but need to employ a deep strategy to reach higher-order thinking skills (Lin et al., 2023b; Pedro et al., 2019). The question is, however, what kind of higher-order thinking is most needed for humans to adapt to the AI era? In this regard, Authors (2024b)'s categorization of AI thinking may provide valuable references. This study defined AI thinking as the high-order thinking that every citizen should possess to adapt to the future AI-powered society. Additionally, the four most critical types of higher-order thinking were identified, including engineering thinking, design thinking, computational thinking, and systems thinking. Among them, design thinking and engineering thinking are logically related, with the former referring to "designing conceived results that do not exist in the real world" and the latter referring to "transforming the design into real artifacts". Therefore, design thinking and engineering thinking are frequently merged into "engineering design thinking" in AI education, which requires students not only to identify problems and propose creative solutions based on real-world needs, but also to develop, test, and optimize AI systems based on these solutions. Computational thinking can enable students to comprehend AI problem-solving mechanisms, which can be fostered through seven phases in AI education: decomposition and modularity, abstraction and modeling, data and

labeling, training and simulation, deployment and inference, optimization and iteration, reuse and transfer. Systems thinking penetrates into engineering design thinking and computational thinking, which work together in the entire process of designing, developing, testing and optimizing AI system. Students are required to reasonably coordinate various elements such as system function, structure, and aesthetics, as well as the interactions between them. Therefore, educators should pay more attention to examining the learning effects of AI thinking through matched learning designs.

4.3. Future challenges of K-12 AI education

4.3.1. Inclusive attention to underrepresented groups

AI is typically regarded as a computer science education offered at the post-secondary level and mostly to high-achieving male students, leaving girls and less able students underrepresented (Delaine et al., 2016). Therefore, inclusive attention was emphasized in the review studies to ensure the success of AI education. For example, Chiu et al. (2021) claimed that further research should investigate how the curriculum addresses the needs of varied academic abilities and gender differences. Besides, Hsu et al. (2022) deemed that further research should explore the effects of different methods on gender-specific AI learning. Similarly, Kim et al. (2023) emphasized that future research will examine the disparities in naive conceptions between genders. Moreover, Sanusi et al. (2023a) suggested that future work could consider how to best introduce ML to middle schools based on the uniqueness of each social context.

Future efforts are required to explore the reasons for the imbalance and how to achieve inclusive AI education. On the one hand, several studies have found that girls have less interest, competence, and confidence in their discipline skills due to the stereotypes that boys are better at computer science activities (Master et al., 2017). Thus, capturing students' interest early at the K-12 level may motivate more underrepresented groups to pursue computer science-related studies or careers (Delaine et al., 2016). On the other hand, most computer science-related activities in K-12 education are designed for high achievers to join competitions in informal setting. The heavy emphasis on coding activities may be too challenging for students who do not have solid foundation such as logical and mathematics thinking (Xia et al., 2022). Since all children should be supported to achieve their potential, K-12 AI teachers should ensure that learning activities are designed and delivered appropriately to students' needs and capabilities (Chiu & Chai, 2020). In addition, more pairing strategies, such as "boy-girl" or "veteran-novice" pairings, could be employed to narrow the gender or foundation gap in AI learning activities.

4.3.2. The dearth of qualified and certified AI teachers

Nowadays, most AI teachers did not receive formal AI education during their studies (Chiu & Chai, 2020). A large number of in-service K-12 teachers are unable to teach AI courses in schools due to a lack of AI knowledge, skills and confidence. Therefore, the dearth of qualified and certified AI teachers was regarded as a major barrier to implementing K-12 AI education, which is also emphasized in the review studies. For example, Lin et al. (2023a) identified the need to train teachers in designing AR resources, authentic contextual dilemmas, role-play scenarios, and case-based visual stimulation. Additionally, Salas-Pilco (2020) showed that teachers should receive technical training on AI and robotics as well as pedagogical training on how to integrate AI and robotics into various disciplines. Moreover, Xia et al. (2022) contended that future research could investigate how to develop teachers' professional capacities to address inclusion and diversity in their curriculum designs.

Future efforts are required to better prepare teachers for teaching AI courses. In particular, researchers and educator should further collaborate to focus on what constitutes and how to develop teachers' AI teaching competence. For the former, several studies have identified self-efficacy as one of the most important constructs in teaching competency (Burić & Kim, 2020; Orakci et al., 2020). Besides, Sun et al. (2023a) utilized the TPACK framework to identify the teaching competencies required for teaching AI courses at the K-12 level, including pedagogical knowledge, content knowledge, and technological knowledge. Still, all these elements need to be sorted out into a systematic framework for teachers' AI teaching competence. For the latter, Lin and Van Brummelen (2021) worked with non-computer science K-12 teachers via a two-day workshop to co-design integrated AI curriculum to prepare AI teachers. Besides, Kandhofer et al. (2019) developed a professional training and certifying system for teachers in AI education. However, short-term teacher training programs primarily provide teachers with basic AI knowledge and experience without in-depth professional development. Therefore, future research needs to develop long-term curriculum and accompanying tools for teacher training.

5. Conclusion

This study conducted a systematic review for empirical research on K-12 AI education over the past decade with 45 empirical studies. While earlier systematic reviews explored the development of AI education, they provided much more descriptive information than in-depth analysis and meaningful synthesis.

From a research perspective, this study reveals the relationship between publication year, sample size, learning stage, educational setting, research method, research focus and duration, which reveals the following research landscape in K-12 AI education: (1) empirical research on formal K-12 AI education has increased rapidly since 2021, (2) most experiments are conducted in junior middle schools with relatively short durations and large sample sizes, (3) the most popular research method is quasi-experiment, which typically requires larger samples and longer duration, and (4) most studies focus on pedagogical development through quasi-experiments or case studies in primary schools. Concerning the future research, some important topics are discussed as follows: (1) K-12 schools should consider AI courses as independent courses, (2) more large-sample and longitudinally designed experiments on AI education in high schools should be conducted, (3) the authentic DBR with extended duration should be conducted, and (4) research on how to evaluate K-12 AI education should be conducted.

From an instruction perspective, this study reveals the relationship between learning stage, pedagogical strategy, learning tool, learning activity, learning content, assessment method, and learning effect of the reviewed papers, which presents the following instructional landscape of K-12 AI education: (1) the overall difficulty of K-12 AI learning activity is still at the introductory level, (2) K-12 AI education prioritizes experiential learning of complex contents, (3) the most widely adopted pedagogical strategies is collaborative learning, which was employed concurrently with project-based learning in the higher grades, (4) the most commonly adopted learning tools are intelligent agents, which are used for advanced learning activities in the higher grades, and (5) the most popular assessments are qualitative methods, which are mostly used to assess students' AI thinking in the lower grades. Concerning the future instruction, some important directions are discussed as follows: (1) educators should consider the cognitive development of each stage to design longer-term and cross-age AI curriculum, (2) educators should prioritize engaging students in physical manipulation of tangible artifacts and sensory-motor activities to facilitate their comprehension of complex AI principles, (3) educators can further combine these pedagogical strategies based on different contents to create an engaging learning environment, (4) more tools with low floors and high ceilings are required to make it easy for novices to get started and potentially work on increasingly complex projects, (5) educators should employ intelligent assessment methods to evaluate and triangulate students' learning performance, and (5) educators should pay more attention to examining the learning effects of AI thinking through matched learning designs.

Moreover, this study summarizes the recommendations for research and instruction. The former is divided into the following four themes: time allocation, samples selection, longitudinal design, rigorous methodology and technical democracy, while the latter is divided into the following four themes: group learning, authentic context, teacher involvement, triangular evidence and learning scaffolding. Meanwhile, future research and instruction should focus on two valuable topics: "inclusive attention to underrepresented groups" and "the dearth of qualified and certified AI teachers".

Overall, this study describes a promising picture of K-12 AI education. It is hoped that this study will be valuable for both practical and research-oriented perspectives on an unexplored but important topic. However, this review also has some limitations. First, there are more empirical studies on applying AI in K-12 education, but fewer empirical studies on K-12 AI education, which results in a relatively small sample size. Second, we only reviewed papers included in the WOS and Scopus, which ensured a representative sample but lacking comprehensiveness. Third, we did not consider social and economic factors when analyzing the reviewed papers, which could affect the implementation of K-12 AI education due to the widespread digital divide (Su et al., 2023b). Therefore, future research will expand the sample size or range, as well as incorporate social and economic factors to fully explore the research and instruction of K-12 AI education.

Credit authorship statement

Xiaofan Liu: Writing - original draft, Methodology, Writing - review & editing. Baichang Zhong: Conceptualization, Writing - review & editing, Supervision.

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Declaration of competing interest

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Data availability

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.edurev.2024.100642>.

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